A Project Report on

Event participation data analysis

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Submitted

By

24B11AI204 – Kowsalya teja

24B11AI303 – N.sandhya

24B11AI397 – Shreya Bishal

Under the esteemed guidance of

Mrs. K.Sowjaaya

Assistant Professor

DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

ADITYA UNIVERSITY

(Formerly Aditya Engineering College (A))

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ABSTRACT

In today's data-driven world, large volumes of event-related data are generated through registrations and participation records. However, this data often remains underutilized due to the lack of proper analysis techniques. This project focuses on applying data mining methods to analyze and predict event participation patterns, helping organizers make informed decisions.

The dataset used in this project contains details such as event ID, event name, location, type of event, and number of participants. The data is first preprocessed to handle missing values, remove duplicates, and convert it into a structured format suitable for analysis. Feature engineering is then applied to extract meaningful attributes that influence participation.

For analysis, clustering is performed using K-Means Clustering, which groups events into categories such as low, medium, and high participation based on attendance. This helps in identifying patterns and understanding which types of events attract more participants. In addition, classification is carried out using Decision Tree, which predicts future event participation levels based on historical data.

The project also uses various data visualization techniques, including bar charts, pie charts, and scatter plots, to present the results in a clear and understandable manner. These visualizations make it easier to identify trends, compare event performance, and interpret model outcomes.

The performance of the predictive model is evaluated by comparing predicted results with actual data, achieving a satisfactory level of accuracy depending on data quality and feature selection. The overall results demonstrate that applying data mining techniques can significantly improve the understanding of event participation behavior.

In conclusion, this project highlights the importance of data mining in event management and provides a practical approach to analyzing and predicting participation, which can help organizers plan better and improve event success rates

Index

Chapter .No	Table of Content	Page No.
1	**Introduction**	6
	1.1 Introduction	6
	1.2 Problem Statement	6
	1.3 Objectives of the Project	6
	1.4 Scope of the Project	7
2	**System Overview**	7
	2.1 Existing System	7
	2.2 Limitations of Existing System	8
	2.3 Proposed System	8
	2.4 Software Requirements	8
	2.5 Hardware Requirements	9
3	**Technologies Used**	9
	3.1 Python & Google Colab	9
	3.2 Machine Learning Algorithms	9
	3.3 Libraries & Tools	10
4	**System Architecture**	10
5	**Data Preprocessing**	12
	5.1 Sample data	12
	5.2 Data Set Preprocessing	12
6	**System Modules**	13
	6.1 Data Preprocessing Modules	13
	6.2 Feature Selection Module	13

	6.3 Training Module	13
	6.4 Prediction Module	14
	6.5 Result Display Module	14
7	**Implementation**	14
	7.1 Data Connection	15
	7.2 Input Validation	15
	7.3 Exception Handling	15
	7.4 Testing with new patient data	15
	7.5 Training and evaluating model	16
8	**Output Screens**	16
	8.1 Data upload screen	16
	8.2 Prediction Screen	17
	8.3 Roc curve	17
	8.4 Graphs	18
9	**Conclusion**	19
10	**Future Scope**	19
11	**References**	20

LIST OF FIGURES

S.No	Figure	Page No
1	Figure 5.1 : Sample Data	12
2	Figure 7.1: Decision Tree model trained on patient data	16
3	Figure 8.1: Event data uploaded and prediction result	17
4	Figure 8.2: Event participation analysus and prediction	17
5	Figure 8.3: ROC curve	18
6	Figure 8.4: Graohs and visualization results	18

1: INTRODUCTION

1.1 Introduction

In today's digital era, a large amount of data is generated from event registrations and participation. However, this data is often not properly analyzed to extract meaningful insights. Understanding participant behavior and event performance is important for improving future events. Data mining techniques help in analyzing large datasets and identifying hidden patterns that support better decision-making.

This project focuses on analyzing event participation data using techniques like **K-Means Clustering** to group events based on participation levels and **Decision Tree** to predict future event success. The results are presented using visualizations such as graphs and charts, making it easier to understand trends and improve event planning strategies

1.2 Problem Statement

Traditional event analysis methods rely on manual observation or basic statistical techniques.

• Cannot efficiently analyze large and diverse event participation datasets
• Fail to identify participation patterns and trends clearly
• Lack the ability to predict future event performance
• Provide limited insights into factors affecting event success

The challenge addressed by this project is to develop an intelligent data mining system that integrates data preprocessing, feature selection, and machine learning techniques such as **K-Means Clustering** and **Decision Tree** to deliver accurate, scalable, and meaningful analysis of event participation.

1.3 Objectives of the Project

- Develop a data mining system for analyzing and predicting event participation using event data.

• Perform data preprocessing to handle missing values, inconsistencies, and duplicates
• Identify key factors such as event type, location, and participant count influencing attendance
• Apply and analyze **K-Means Clustering** to group events based on participation levels
• Apply **Decision Tree** to predict event performance
• Evaluate model performance using accuracy and confusion matrix
• Provide clear and structured insights through graphs and visualizations
• Ensure the system is scalable and useful for real-world event management

1.4 Scope of the Project

• The project focuses on analyzing event participation data to extract meaningful insights
• It covers data preprocessing, feature selection, and data mining techniques
• It uses **K-Means Clustering** to group events based on participation levels
• It applies **Decision Tree** to predict future event performance
• It includes visualization of data using graphs and charts for better understanding
• The system can be extended to real-time event data and larger datasets
• It can be used by event organizers to improve planning and decision-making

2: SYSTEM OVERVIEW

2.1 Existing System

In the current event management system, data analysis is mostly done manually or using basic tools like spreadsheets. Event organizers rely on simple observations and reports, which makes it difficult to analyze large datasets efficiently. Important patterns in participation are often missed, and decision-making is not fully data-driven.

2.2 Limitations of Existing System

- Manual analysis is time-consuming and error-prone
  • Cannot handle large and diverse event datasets efficiently
  • Difficult to identify participation patterns and trends
  • Lack of prediction for future event performance
  • Limited visualization and reporting capabilities
  • Not suitable for real-time analysis

2.3 Proposed System

The proposed system uses data mining techniques to automate event participation analysis and prediction. Key features include:

• Data preprocessing to clean the dataset, handle missing values, and remove duplicates
• Feature selection to identify important factors such as event type, location, and participants
• Use of machine learning algorithms like **K-Means Clustering** and **Decision Tree**
• Grouping events into Low, Medium, and High participation levels
• Predicting future event performance
• Providing clear insights using graphs and visualizations
• Scalable and useful for real-world event management

2.4 Software Requirements

- **Python** programming language for coding
- **Google Colab** for running and testing machine learning models
- Libraries: **pandas, numpy, matplotlib, scikit-learn, seaborn**
- Operating System: **Windows 10 or above**
- Browser for Google Colab (e.g., Chrome, Edge)

2.5 Hardware Requirements

- Processor: **Intel i3 or higher**
- RAM: **4 GB or more**
- Hard Disk: **10 GB free space**
- Internet connection (required for Google Colab)
- Monitor and keyboard

3: TECHNOLOGIES USED

3.1 Python & Google Colab

• Python:
Python is used as the main programming language for this project due to its simplicity and powerful libraries for data analysis and machine learning. It provides support for data preprocessing, visualization, and implementation of algorithms like clustering and classification.

• Google Colab:
Google Colab is a cloud-based platform used to write and execute Python code. It allows easy access without installation, supports required libraries, and provides a convenient environment for running data mining and machine learning models.

3.2 Machine Learning Algorithms

To analyze and predict event participation, the following machine learning algorithms were applied:

• **K-Means Clustering**:
Used to group events into clusters such as Low, Medium, and High participation based on attendee data. It helps in identifying hidden patterns in event performance.

• **Decision Tree**:
Used for classification and prediction of event participation levels. It builds a tree-like model that is easy to understand and interpret.

These algorithms were selected to effectively analyze participation patterns and predict future event outcomes with good accuracy.

3.3 Libraries & Tools

Several Python libraries and tools were used to implement the project:

- **pandas:** For data manipulation and analysis
- **numpy:** For numerical calculations
- **scikit-learn:** For implementing machine learning algorithms
- **matplotlib & seaborn:** For visualizing data and results
- **Google Colab:** For writing, running, and sharing Python code online

4: SYSTEM ARCHITECTURE

The system architecture of this project represents the overall workflow for analyzing event participation using data mining techniques. It consists of multiple stages that transform raw event data into meaningful insights through a structured process.

The first stage is Data Collection, where event-related data is gathered from reliable sources. The dataset includes attributes such as Event ID, Event Name, Location, Event Type, Date, and number of Participants.

The next stage is Data Integration, where all collected data is combined into a single structured dataset. This ensures that all event-related information is available for comprehensive analysis.

The third stage is Data Preprocessing, which is performed using Python tools. In this stage, missing values are handled, duplicate records are removed, and inconsistent

data is corrected. Data transformation and encoding are also applied to prepare the dataset for analysis.

After preprocessing, the system moves to the Data Visualization stage. Various charts such as bar graphs, pie charts, and scatter plots are used to understand participation trends, identify popular events, and detect patterns.

The next stage is Data Mining, where machine learning algorithms are applied:
• K-Means Clustering to group events into Low, Medium, and High participation
• Decision Tree to predict future event performance

Following this, the Model Evaluation stage is carried out. The performance of the model is evaluated using accuracy and confusion matrix to ensure reliable predictions.

Finally, the system produces Results and Insights, where meaningful conclusions are drawn. The system identifies patterns in participation and provides insights to help event organizers make better decisions.

[Data Collection]

↓

[Data Integration]

↓

[Data Preprocessing]

↓

[Data Visualization]

↓

[Data Mining]

(Clustering + Classification)

↓

[Model Evaluation]

↓

[Results & Insights]

5: DATA PREPROCESSING

5.1 Sample Data / Records

Event_ID	Event_Name	Location	Date_Time	Attendee_Name	Attendee_Email	Attendee_Phone
E0001	Orientation Progra	Vijayawada	27-09-2024 19:45	Attendee_0001	attendee0001@example.com	9523734972
E0002	Sports Meet	Kakinada	11-04-2025 15:30	Attendee_0002	attendee0002@example.com	9998717130
E0003	Webinar on Data M	Visakhapatnam	09-04-2024 16:45	Attendee_0003	attendee0003@example.com	9748061058
E0004	Alumni Meet	Visakhapatnam	09-12-2024 14:15	Attendee_0004	attendee0004@example.com	9880932287
E0005	Sports Meet	Surampalem	04-04-2025 18:15	Attendee_0005	attendee0005@example.com	9130349564
E0006	Coding Contest	Surampalem	19-04-2025 19:30	Attendee_0006	attendee0006@example.com	9699121577
E0007	Cultural Fest	Vijayawada	20-07-2025 12:45	Attendee_0007	attendee0007@example.com	9299502978
E0008	Sports Meet	Rajahmundry	08-10-2025 13:00	Attendee_0008	attendee0008@example.com	9379394470
E0009	Hackathon	Vijayawada	23-08-2025 10:15	Attendee_0009	attendee0009@example.com	9334564360
E0010	Seminar on ML	Visakhapatnam	05-12-2024 15:45	Attendee_0010	attendee0010@example.com	9457319431

Figure 5.1 : Sample Data

Note: The participation level is categorized as:

0 = Low Participation

1 = Medium Participation

2 = High Participation

5.2 Dataset Preprocessing

Before applying machine learning algorithms, the dataset undergoes preprocessing to improve model performance:

• Handling Missing Values: Missing data is filled or removed to ensure dataset completeness

• Data Cleaning: Duplicate records and inconsistent entries are removed

• Normalization / Scaling: Numeric values like participant count are scaled to improve model accuracy

• Encoding Categorical Data: Text fields such as Event Type and Location are converted into numeric format

• Feature Selection: Important attributes are selected for analysis

These steps ensure the dataset is clean, consistent, and ready for applying K-Means Clustering and Decision Tree for clustering and prediction.

6: SYSTEM MODULES

The proposed event participation analysis system is divided into major functional modules. Each module performs specific tasks to ensure efficient data processing, accurate analysis, and meaningful insights.

6.1 Data Preprocessing Module

Purpose: Prepare the raw dataset for analysis

• **Functions:**

o Handle missing values (fill or remove incomplete records)

o Remove duplicate and inconsistent data

o Convert categorical data (Event Type, Location) into numeric form

o Normalize/scale numerical values like Participants

• **Outcome:** A clean and structured dataset ready for analysis

6.2 Feature Selection Module

• **Purpose:** Identify important factors influencing event participation

• **Key Features Used:**

o Event Type

o Location

o Date

o Number of Participants

• **Technique:** Selection of relevant attributes to improve analysis

• **Outcome:** Reduced dataset with meaningful features

6.3 Training Module

Purpose: Train models for clustering and prediction

• **Algorithms Used:**

o K-Means Clustering – Groups events into Low, Medium, High participation

o Decision Tree – Predicts event performance

• **Process:**

o Split dataset into training and testing sets

o Train models on the dataset

o Adjust parameters for better accuracy

• Outcome: Trained models for analysis and prediction

6.4 Prediction Module

• **Purpose:** Predict participation level of events

• **Process:**

o Input event details into the trained model

o Classify event into Low, Medium, or High participation

o Store prediction results

• **Outcome:** Each event is categorized with a predicted participation level

6.5 Result Display Module

• **Purpose:** Present results in an understandable format

• **Features:**
o Display event participation levels
o Show graphs like bar charts, pie charts, scatter plots
o Provide insights for decision-making
o Option to export results

• **Outcome:** Clear and user-friendly visualization of results

7: IMPLEMENTATION

The implementation of the Event Participation Analysis System involves loading event data, validating inputs, handling errors, and analyzing participation using machine learning techniques. The system is developed using Python in Google Colab with models like clustering and classification.

7.1 Data Connection / Database

• **Purpose:** Load event data for analysis

• **Implementation:**

o Event records are stored in an Excel/CSV file

o The dataset is loaded using Python libraries like pandas

o Data is prepared for preprocessing and analysis

Outcome: Event data is successfully loaded and ready for processing

7.2 Input Validation

• **Purpose:** Ensure that the event data is correct and usable

• **Implementation:**

o Validate numeric fields like number of participants

o Check categorical fields like Event Type and Location

o Handle missing or incomplete records

Outcome: Clean and valid data is used for analysis

7.3 Exception Handling

Purpose: Prevent system errors and ensure smooth execution

Implementation:

o Handle missing dataset errors

o Manage invalid data inputs

o Catch runtime errors during processing

Outcome: System runs reliably without crashing

.

7.4 Testing with New Patient Data

Purpose: Analyze and predict participation for new events

Implementation:

o Input new event details into the system

o Apply trained models to classify participation level

o Categorize events as Low, Medium, or High participation

Outcome: New events can be analyzed effectively

7.5 .Training and evaluating model

```
# Train Model
model = DecisionTreeClassifier(random_state=42)
model.fit(X_train, y_train)

# Prediction
y_pred = model.predict(X_test)

# Evaluation
accuracy = accuracy_score(y_test, y_pred)
print("Accuracy:", accuracy * 100)

print("\nConfusion Matrix:")
print(confusion_matrix(y_test, y_pred))
```

```
Accuracy: 100.0
```

Figure 7.1: Decision Tree model trained on patient data

8: OUTPUT SCREENS

The Event Participation Analysis System provides a user-friendly interface for users to upload event data, view analysis results, and understand participation trends. The main output screens are described below:

8.1 Data Upload Screen

Purpose: Allow users to upload event data for analysis

• **Functionality:**

o Users can upload a CSV/Excel file containing event records

o The system reads and displays the dataset

```
Choose Files event_parti...00 data.xlsx
event_participation_500 data.xlsx(application/vnd.openxmlformats-officedocument.spreadsheetml.sheet) - 42441 bytes, last modified: 3/23/2026 - 100% done
Saving event_participation_500 data.xlsx to event_participation_500 data.xlsx
```

Figure 8.1: Event data uploaded for analysis

• **Outcome:** Data is successfully loaded and ready for preprocessing and analysis

8.2 Prediction Screen

Purpose: Display clustering and prediction results

• **Functionality:**

o Shows event participation levels (Low, Medium, High)

o Displays event details such as Event Type, Location, and Participants

o Applies K-Means Clustering and Decision Tree

```
Accuracy: 100.0

Confusion Matrix:
 [[100]]
/usr/local/lib/python3.12/dist-packages/sklearn/metrics/_classification.py:407: UserWarning: A single label was found in 'y_true' and 'y_pre
  warnings.warn(
```

Figure 8.2: Event participation analysis and prediction

• **Outcome:** Users can clearly see how events are grouped and predicted

8.3. Roc curve for decision tree

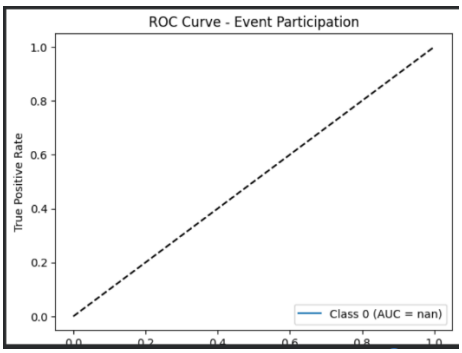

Figure 8.4: Roc curve

8.4. Graphs / Visualization Screen

Purpose: Display visual representation of data

• **Functionality:**

o Bar charts showing cluster distribution

o Pie charts showing percentage of participation levels

o Scatter plots for clustering visualization

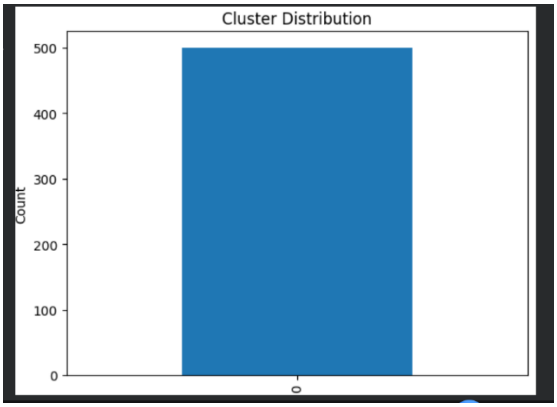

Figure 8.4: Graphs and visualization results

• **Outcome:** Helps users easily understand trends and patterns in event participation

9: CONCLUSION

The Event Participation Analysis project successfully developed a data mining-based system to analyze and predict event participation using event data.

Key points:

• The system accepts event datasets and performs data preprocessing, including cleaning, handling missing values, and removing duplicates

• Important features such as Event Type, Location, and Participants were selected for accurate analysis

• **K-Means Clustering** was used to group events into Low, Medium, and High participation levels

• **Decision Tree** was applied to predict event performance

• Input validation and error handling ensure reliability and smooth execution

• The system generates structured reports and visualizations for better understanding

Outcome:

The project demonstrates how data mining techniques can help in analyzing

participation patterns, predicting event success, and supporting better decision-making for event management.

10: Future Scope

Several improvements can be made to extend the functionality and usability of the system:

1. **Interactive GUI:** Develop a user-friendly interface for easy data input and visualization of results
2. **Web-Based Application:** Deploy the system online so event organizers can access it anytime
3. **Support for Larger Datasets:** Enhance the system to handle real-time and large-scale event data
4. **Additional Algorithms:** Integrate advanced models like Random Forest or Neural Networks to improve prediction accuracy beyond **Decision Tree**
5. **Improved Clustering Methods:** Use advanced clustering techniques beyond **K-Means Clustering** for better grouping
6. **Automated Reports:** Generate PDF or Excel reports automatically for event analysis
7. **Integration with Event Platforms:** Connect the system with online event platforms for real-time data collection

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